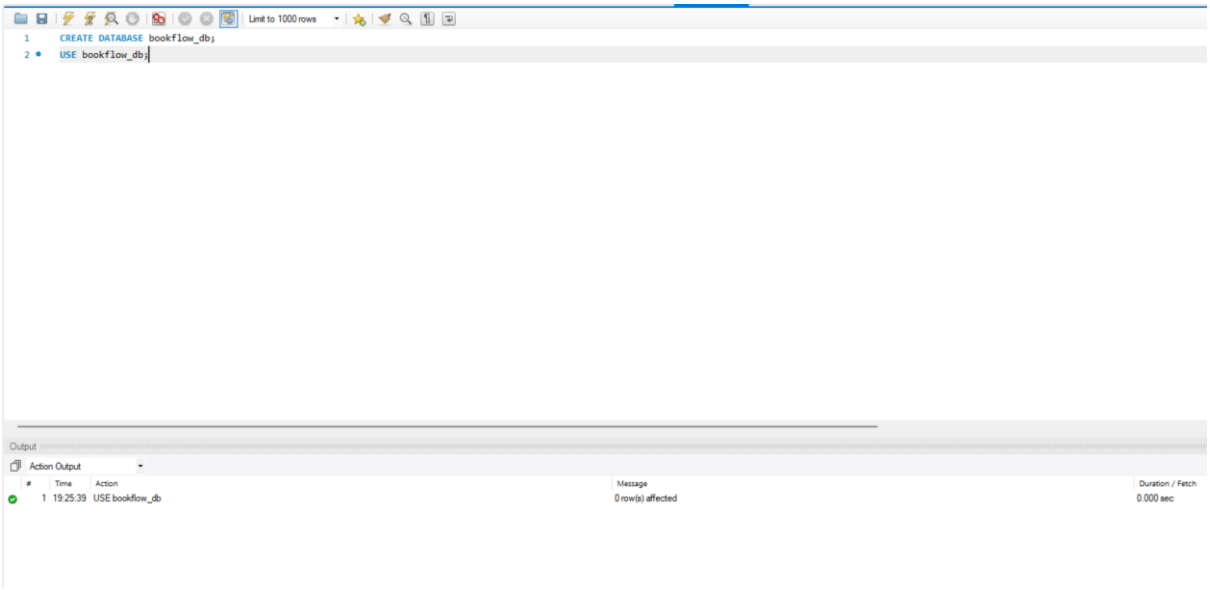
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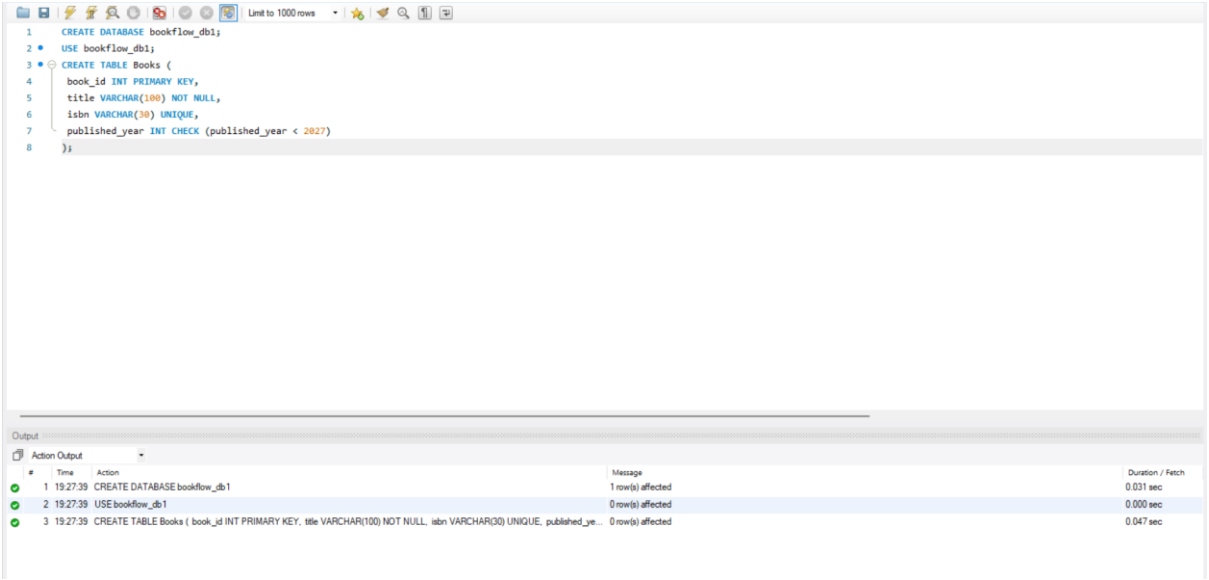
K.Poojitha Sai

Week-2

Step 1 — Create Database



Step 2 — Create Books Table (DDL + Constraints)



Step 3 — Insert Books

```
    isbn VARCHAR(30) UNIQUE,
    published_year INT CHECK (published_year < 2027)
    );
INSERT INTO Books (book_id, title, isbn, published_year) VALUES
    (1, 'The Great Gatsby', '9780743273565', 1925),
    (2, 'To Kill a Mockingbird', '9780061120084', 1960),
    (3, '1984', '9780451524935', 1949);
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	19:27:39	CREATE DATABASE bookflow_db1	1 row(s) affected	0.031 sec
2	19:27:39	USE bookflow_db1	0 row(s) affected	0.000 sec
3	19:27:39	CREATE TABLE Books (book_id INT PRIMARY KEY, title VARCHAR(100) NOT NULL, isbn VARCHAR(30) UNIQUE, published_ye...	0 row(s) affected	0.047 sec
4	19:28:59	INSERT INTO Books (book_id, title, isbn, published_year) VALUES (1, 'The Great Gatsby', '9780743273565', 1925), (2, 'To Kill a Mocki...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.032 sec

Step 4 -Display books

```
CREATE DATABASE bookflow_db1;
USE bookflow_db1;
CREATE TABLE Books (
    book_id INT PRIMARY KEY,
    title VARCHAR(100) NOT NULL,
    isbn VARCHAR(30) UNIQUE,
    published_year INT CHECK (published_year < 2027)
    );
INSERT INTO Books (book_id, title, isbn, published_year) VALUES
    (1, 'The Great Gatsby', '9780743273565', 1925),
    (2, 'To Kill a Mockingbird', '9780061120084', 1960),
    (3, '1984', '9780451524935', 1949);
SELECT * FROM Books
```

Result Grid

book_id	title	isbn	published_year
1	The Great Gatsby	9780743273565	1925
2	To Kill a Mockingbird	9780061120084	1960
3	1984	9780451524935	1949
NULL	NULL	NULL	NULL

Books 1

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	19:27:39	CREATE DATABASE bookflow_db1	1 row(s) affected	0.031 sec
2	19:27:39	USE bookflow_db1	0 row(s) affected	0.000 sec
3	19:27:39	CREATE TABLE Books (book_id INT PRIMARY KEY, title VARCHAR(100) NOT NULL, isbn VARCHAR(30) UNIQUE, published_y...	0 row(s) affected	0.047 sec
4	19:28:59	INSERT INTO Books (book_id, title, isbn, published_year) VALUES (1, 'The Great Gatsby', '9780743273565', 1925), (2, 'To Kill a Moc...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.032 sec
5	19:30:05	SELECT * FROM Books LIMIT 0, 1000	3 row(s) returned	0.016 sec / 0.000 sec

```
    );

    INSERT INTO Members (member_id, full_name, email) VALUES
    (101, 'Poojitha Sai', 'poojithasai@email.com'),
    (102, 'Emma Wilson', 'emma.wilson@email.com'),
    (103, 'Michael Brown', 'michael.brown@email.com');

    SELECT * FROM Members;

    CREATE TABLE Loans (
```

Result Grid

member_id	full_name	email
101	Poojitha Sai	poojithasai@email.com
102	Emma Wilson	emma.wilson@email.com
103	Michael Brown	michael.brown@email.com
NULL	NULL	NULL

Members 2

Output

Action Output

#	Time	Action	Message
3	13:48:05	USE bookflow_db1	0 row(s) affected
4	13:48:10	CREATE TABLE Books (book_id INT PRIMARY KEY, title VARCHAR(100) NOT NULL, isbn VARCHAR(30) UNIQUE, pu...	0 row(s) affected
5	13:48:15	INSERT INTO Books (book_id, title, isbn, published_year) VALUES (1, 'The Great Gatsby', '9780743273565', 1925), (2, 'To Kill a Mo...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
6	13:48:18	SELECT * FROM Books LIMIT 0, 1000	3 row(s) returned
7	13:48:24	CREATE TABLE Members (member_id INT PRIMARY KEY, full_name VARCHAR(100), email VARCHAR(100) UNIQUE)	0 row(s) affected
8	13:48:28	INSERT INTO Members (member_id, full_name, email) VALUES (101, 'Poojitha Sai', 'poojithasai@email.com'), (102, 'Emma Wilson', '...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
9	13:48:33	SELECT * FROM Members LIMIT 0, 1000	3 row(s) returned

```sql
CREATE TABLE Loans (
 loan_id INT PRIMARY KEY,
 member_id INT,
 book_id INT,
 loan_date DATE,
 FOREIGN KEY (member_id) REFERENCES Members(member_id),
 FOREIGN KEY (book_id) REFERENCES Books(book_id)
 );
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
4	19:28:59	INSERT INTO Books (book_id, title, isbn, published_year) VALUES (1, 'The Great Gatsby', '9780743273565', 1925), (2, 'To Kill a Mo...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.032 sec
5	19:30:05	SELECT * FROM Books LIMIT 0, 1000	3 row(s) returned	0.016 sec / 0.000 sec
6	19:33:28	CREATE TABLE Members (member_id INT PRIMARY KEY, full_name VARCHAR(100), email VARCHAR(100) UNIQUE)	0 row(s) affected	0.110 sec
7	19:33:35	INSERT INTO Members (member_id, full_name, email) VALUES (101, 'John Smith', 'john.smith@email.com'), (102, 'Emma Wilson', 'e...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.032 sec
8	19:33:35	SELECT * FROM Members1 LIMIT 0, 1000	Error Code: 1146. Table 'bookflow_db1.members1' doesn't exist	0.000 sec
9	19:33:51	SELECT * FROM Members LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
10	19:34:27	CREATE TABLE Loans (loan_id INT PRIMARY KEY, member_id INT, book_id INT, loan_date DATE, FOREIGN KEY (member_i...	0 row(s) affected	0.125 sec

```sql
 );

INSERT INTO Loans (loan_id, member_id, book_id, loan_date) VALUES
(1, 101, 1, '2025-01-05'), (2, 102, 2, '2025-01-08'),
(3, 103, 3, '2025-01-10'), (4, 101, 2, '2025-02-01'),
(5, 102, 1, '2025-02-05'), (6, 103, 2, '2025-02-12'),
(7, 101, 3, '2025-03-01'), (8, 102, 3, '2025-03-07'),
(9, 103, 1, '2025-03-15'), (10, 101, 1, '2025-04-01');
SELECT * FROM Loans;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Call Content:

loan_id	member_id	book_id	loan_date
1	101	1	2025-01-05
2	102	2	2025-01-08
3	103	3	2025-01-10
4	101	2	2025-02-01
5	102	1	2025-02-05

Loans 3 | Apply | Revert

Output

Action Output

#	Time	Action	Message	Duration / Fetch
6	19:33:28	CREATE TABLE Members (member_id INT PRIMARY KEY, full_name VARCHAR(100), email VARCHAR(100) UNIQUE)	0 row(s) affected	0.110 sec
7	19:33:35	INSERT INTO Members (member_id, full_name, email) VALUES (101, 'John Smith', 'john.smith@email.com'), (102, 'Emma Wilson', 'e...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.032 sec
8	19:33:35	SELECT * FROM Members1 LIMIT 0, 1000	Error Code: 1146. Table 'bookflow_db1.members1' doesn't exist	0.000 sec
9	19:33:51	SELECT * FROM Members LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
10	19:34:27	CREATE TABLE Loans (loan_id INT PRIMARY KEY, member_id INT, book_id INT, loan_date DATE, FOREIGN KEY (member_i...	0 row(s) affected	0.125 sec
11	19:35:19	INSERT INTO Loans (loan_id, member_id, book_id, loan_date) VALUES (1, 101, 1, '2025-01-05'), (2, 102, 2, '2025-01-08'), (3, 103, ...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.047 sec
12	19:35:19	SELECT * FROM Loans LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec

```sql
SELECT * FROM Loans;

SELECT m.full_name AS Member_Name, b.title AS Book_Title
FROM Loans l
INNER JOIN Members m ON l.member_id = m.member_id
INNER JOIN Books b ON l.book_id = b.book_id;
```

Result Grid | Filter Rows: | Export: | Wrap Call Content:

Member_Name	Book_Title
John Smith	The Great Gatsby
John Smith	To Kill a Mockingbird
John Smith	1984
John Smith	The Great Gatsby
Emma Wilson	To Kill a Mockingbird

Result 4 | Read Only

Output

Action Output

#	Time	Action	Message	Duration / Fetch
7	19:33:35	INSERT INTO Members (member_id, full_name, email) VALUES (101, 'John Smith', 'john.smith@email.com'), (102, 'Emma Wilson', 'e...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.032 sec
8	19:33:35	SELECT * FROM Members1 LIMIT 0, 1000	Error Code: 1146. Table 'bookflow_db1.members1' doesn't exist	0.000 sec
9	19:33:51	SELECT * FROM Members LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
10	19:34:27	CREATE TABLE Loans (loan_id INT PRIMARY KEY, member_id INT, book_id INT, loan_date DATE, FOREIGN KEY (member_i...	0 row(s) affected	0.125 sec
11	19:35:19	INSERT INTO Loans (loan_id, member_id, book_id, loan_date) VALUES (1, 101, 1, '2025-01-05'), (2, 102, 2, '2025-01-08'), (3, 103, ...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.047 sec
12	19:35:19	SELECT * FROM Loans LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec
13	19:36:00	SELECT m.full_name AS Member_Name, b.title AS Book_Title FROM Loans l INNER JOIN Members m ON l.member_id = m.member...	10 row(s) returned	0.000 sec / 0.000 sec

```sql
SELECT published_year, COUNT(book_id) AS Total_Books
FROM Books
GROUP BY published_year
ORDER BY published_year;
```

Result Grid

published_year	Total_Books
1925	1
1949	1
1960	1

Result 5

Read Only

Output

Action Output

#	Time	Action	Message	Duration / Fetch
8	19:33:35	SELECT * FROM Members1 LIMIT 0, 1000	Error Code: 1146. Table 'bookflow_db1.members1' doesn't exist	0.000 sec
9	19:33:51	SELECT * FROM Members LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
10	19:34:27	CREATE TABLE Loans (loan_id INT PRIMARY KEY, member_id INT, book_id INT, loan_date DATE, FOREIGN KEY (member_i...	0 row(s) affected	0.125 sec
11	19:35:19	INSERT INTO Loans (loan_id, member_id, book_id, loan_date) VALUES (1, 101, 1, '2025-01-05'), (2, 102, 2, '2025-01-08'), (3, 103, ...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.047 sec
12	19:35:19	SELECT * FROM Loans LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec
13	19:36:00	SELECT m.full_name AS Member_Name, b.title AS Book_Title FROM Loans l INNER JOIN Members m ON l.member_id = m.member...	10 row(s) returned	0.000 sec / 0.000 sec
14	19:36:29	SELECT published_year, COUNT(book_id) AS Total_Books FROM Books GROUP BY published_year ORDER BY published_year L...	3 row(s) returned	0.000 sec / 0.000 sec

```sql
ORDER BY published_year;

CREATE TABLE Donation_History (
  donation_id INT PRIMARY KEY,
  book_id INT,
  donor_name VARCHAR(100),
  donation_date DATE,
  FOREIGN KEY (book_id) REFERENCES Books(book_id)
);
START TRANSACTION;
INSERT INTO Books (book_id, title, isbn, published_year)
VALUES (4, 'Animal Farm', '9780451526342', 1945);
INSERT INTO Donation_History (donation_id, book_id, donor_name, donation_date)
VALUES (1, 4, 'Raj Kumar', CURDATE());
COMMIT;
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
13	19:36:00	SELECT m.full_name AS Member_Name, b.title AS Book_Title FROM Loans l INNER JOIN Members m ON l.member_id = m.member...	10 row(s) returned	0.000 sec / 0.000 sec
14	19:36:29	SELECT published_year, COUNT(book_id) AS Total_Books FROM Books GROUP BY published_year ORDER BY published_year L...	3 row(s) returned	0.000 sec / 0.000 sec
15	19:37:17	CREATE TABLE Donation_History (donation_id INT PRIMARY KEY, book_id INT, donor_name VARCHAR(100), donation_date ...	0 row(s) affected	0.093 sec
16	19:37:17	START TRANSACTION	0 row(s) affected	0.000 sec
17	19:37:17	INSERT INTO Books (book_id, title, isbn, published_year) VALUES (4, 'Animal Farm', '9780451526342', 1945)	1 row(s) affected	0.000 sec
18	19:37:17	INSERT INTO Donation_History (donation_id, book_id, donor_name, donation_date) VALUES (1, 4, 'Raj Kumar', CURDATE())	1 row(s) affected	0.000 sec
19	19:37:17	COMMIT	0 row(s) affected	0.000 sec

```sql
CREATE INDEX idx_books_isbn ON Books(isbn);
SELECT * FROM Books WHERE isbn = '9780451524935';
```

Result Grid

book_id	title	isbn	published_year
3	1984	9780451524935	1949
NULL	NULL	NULL	NULL

Books 6

Apply Revert

Output

Action Output

#	Time	Action	Message	Duration / Fetch
15	19:37:17	CREATE TABLE Donation_History (donation_id INT PRIMARY KEY, book_id INT, donor_name VARCHAR(100), donation_date ...	0 row(s) affected	0.093 sec
16	19:37:17	START TRANSACTION	0 row(s) affected	0.000 sec
17	19:37:17	INSERT INTO Books (book_id, title, isbn, published_year) VALUES (4, 'Animal Farm', '9780451526342', 1945)	1 row(s) affected	0.000 sec
18	19:37:17	INSERT INTO Donation_History (donation_id, book_id, donor_name, donation_date) VALUES (1, 4, 'Raj Kumar', CURDATE())	1 row(s) affected	0.000 sec
19	19:37:17	COMMIT	0 row(s) affected	0.000 sec
20	19:37:56	CREATE INDEX idx_books_isbn ON Books(isbn)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.063 sec
21	19:37:56	SELECT * FROM Books WHERE isbn = '9780451524935' LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec